TRANSITPRO APP GUIDE · HOW TO RUN & EVALUATE

TransitPro – Transport Management System

A web-based database application: ERD → MySQL database → web interface (CRUD)

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Program / Batch	PGD – DSAI, Batch XIV	Course	RDBMS – Database Management Systems
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Database	transport_db (MySQL 8)	Technology	MySQL 8 · Python Flask · HTML/CSS

⏱️ **Quickest way to check (Windows):** unzip → double-click **START_HERE.bat** → enter your MySQL root password → the app opens at **http://127.0.0.1:5000**. The database `transport_db` is created in MySQL automatically. Total time: about 3 minutes. **No need to run it?** The ERD, SQL scripts and screenshots in the ZIP can be reviewed directly (Section 1).

1 What is in the ZIP

FILE / FOLDER	CONTENTS
TransitPro_Presentation.pptx	16-slide presentation of the whole project (slide 3 explains this ZIP)
TransitPro_Overview.pdf	One-page visual summary
TransitPro_App_Guide.pdf	This guide
TransitPro_User_Manual.docx	User manual: how to use every screen, with field reference tables
START_HERE.bat	One-click launcher for Windows (installs packages, connects to MySQL, opens browser)
docs/ERD.png	Entity Relationship Diagram (7 entities, 6 one-to-many relationships)
database/01_schema_mysql.sql	DDL: database, 7 tables, PRIMARY / FOREIGN KEYs, UNIQUE, CHECK, NOT NULL, DEFAULT, indexes, view
database/02_seed_data.sql	DML: sample data (INSERT)
database/03_queries_demo.sql	CRUD, INNER/LEFT/RIGHT/SELF JOIN, GROUP BY/HAVING, sub-queries, view, stored procedure, function, triggers, transaction, constraint tests, GRANT/REVOKE
README.md	Full project report: ERD explanation, data dictionary, SQL concepts, architecture
docs/screenshots/	Screenshots of every page of the application
app.py · db.py · config.py · templates/ · static/	Web application source code (Flask + HTML/CSS)

2 Requirements checklist

ASSIGNMENT REQUIREMENT		WHERE TO SEE IT
ERD with entities & relationships	✔	`docs/ERD.png` · app page **ERD & Project** · README §5

Tables	✔	7 tables in 3NF: customers, drivers, vehicles, routes, shipments, payments, maintenance
Primary Keys	✔	AUTO_INCREMENT PK on every table
Foreign Keys	✔	6 FKs with ON DELETE RESTRICT / CASCADE / SET NULL (`01_schema_mysql.sql`)
Constraints	✔	NOT NULL, UNIQUE (incl. composite), CHECK, DEFAULT. Violations are shown in the web app.
Joins	✔	App pages **SQL Reports** and shipment detail (5-table join) · `03_queries_demo.sql` Part B
CRUD (Insert, View, Update, Delete)	✔	Add / Edit / Delete / List forms for all 7 tables
Web forms: enter, update, search, display	✔	Every list page has a search box. Shipments also have a status filter and tracking-number lookup.
Database connectivity (MySQL)	✔	Python **PyMySQL** driver (`db.py`). Badge "MySQL connected" in the app.

3 How to run the application

Requirement: Python 3.10+ (python.org; on Windows tick "Add python.exe to PATH" during install). MySQL 8.0.16+ with Workbench is optional; without it, the app runs on a built-in SQLite copy of the same database.

RECOMMENDED · WINDOWS

A. One click with MySQL

1. Unzip the project.
2. Double-click **START_HERE.bat** (if SmartScreen appears: *More info → Run anyway*)
3. Type the MySQL **root password** and press **Enter**.
4. The browser opens **127.0.0.1:5000**.

The first run creates `transport_db` automatically. Keep the console window open while you use the app.

NO MYSQL INSTALLED

B. Built-in database

1. Double-click **START_HERE.bat**.
2. At the password prompt, just press **Enter**.
3. The app runs on SQLite with the identical schema and data.

The sidebar badge shows "SQLite connected".

MACOS / LINUX / MANUAL

C. Command line

1. Open a terminal in the project folder.
2. Run the commands below.
3. Open **127.0.0.1:5000**.

```
# Option C – manual run (any OS)
pip install -r requirements.txt                # Flask, PyMySQL, cryptography

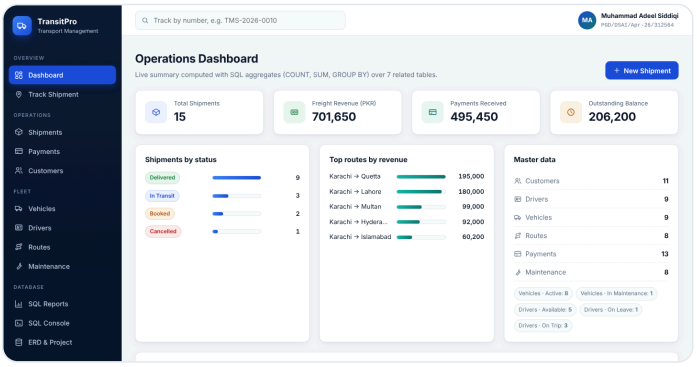
# Windows (Command Prompt)          # macOS / Linux
set MYSQL_PASSWORD=your_password          export MYSQL_PASSWORD=your_password
python app.py                             python3 app.py

# Without MySQL:  set DB_ENGINE=sqlite   (macOS/Linux: export DB_ENGINE=sqlite)
```

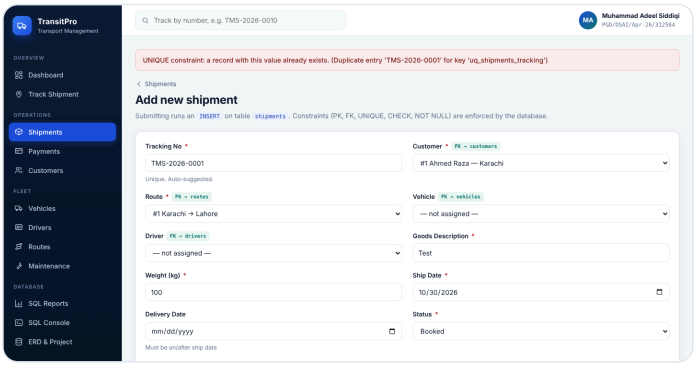
Prefer to create the database yourself? In MySQL Workbench, open and execute `database/01_schema_mysql.sql` , then `02_seed_data.sql` (*File → Open SQL Script → ⚡*). The app then connects to it. Other MySQL settings (user, host, port) are in `config.py` .

4 5-minute evaluation walkthrough

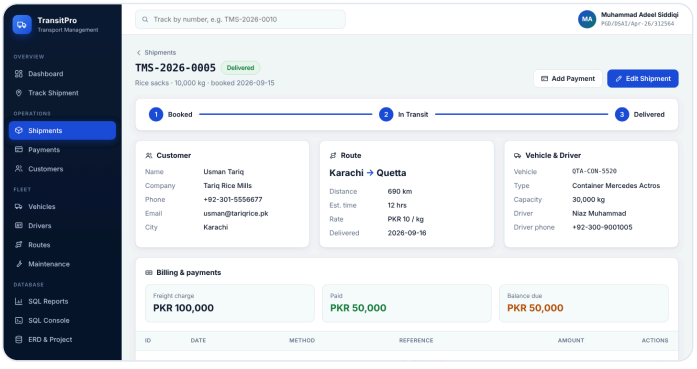
#	ACTION IN THE WEB APP	EXPECTED RESULT (CONCEPT)
1	Open **Dashboard**	KPIs and charts from `COUNT / SUM / GROUP BY` across joined tables
2	**Customers → Add Customer** → Save. Then in Workbench: `SELECT * FROM transport_db.customers;`	**New row appears in MySQL: INSERT and connectivity work**
3	Click ✏️ on a row → change a value → **Update**	**UPDATE: the executed SQL is shown on screen**
4	Type *Karachi* in the search box on **Shipments**	**Filtered list (`WHERE … LIKE`)**
5	Delete customer **#1 Ahmed Raza**	**Refused: FOREIGN KEY with ON DELETE RESTRICT**
6	**Shipments → Add** with tracking no `TMS-2026-0001`	**Refused: UNIQUE constraint**
7	**Shipments → Add** with weight `-5`	**Refused: CHECK constraint**
8	Delete the customer created in step 2	**DELETE succeeds (no related records)**
9	Open any shipment (click its tracking no.)	**5-table INNER/LEFT JOIN: customer, route, vehicle, driver, payments**
10	**SQL Reports** → click each report	**SQL text + live result: INNER, LEFT, SELF JOIN, HAVING, sub-queries**
11	**SQL Console** → run any SELECT	**Ad-hoc read-only queries on the live database**



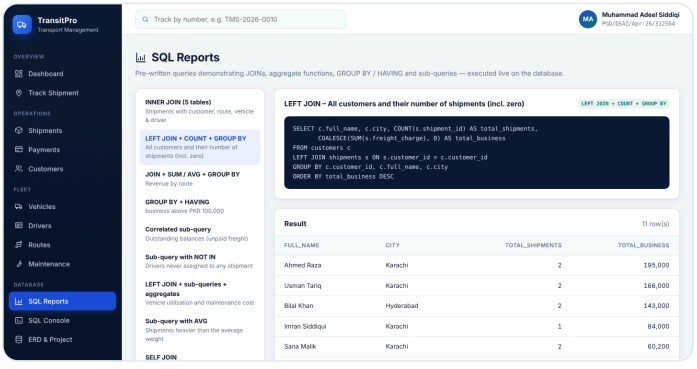
Dashboard (MySQL connected)



Step 6: UNIQUE constraint violation caught and shown



Step 9: shipment detail (multi-table JOIN)



Step 10: SQL Reports (query + live result)

5 Checking the database in MySQL Workbench

- **Tables:** Schemas panel → ⟳ refresh → `transport_db` → Tables (7) → right-click → *Select Rows*.
- **ERD from the live database:** *Database → Reverse Engineer…* → select `transport_db` → Execute → Finish.

- **SQL concepts:** open `database/03_queries_demo.sql` and run each statement with Ctrl + Enter (Parts A–K: CRUD, joins, aggregates, sub-queries, views, procedure, function, triggers, transaction, constraint tests, DCL).
- **Reset to original data:** run `01_schema_mysql.sql` and `02_seed_data.sql` again, or `DROP DATABASE transport_db;` and start the app (it rebuilds the database).

Entity Relationship Diagram



6 Troubleshooting

MESSAGE	FIX
"Python is NOT installed"	Install Python 3.10+ with "Add python.exe to PATH", then run START_HERE again.
"WRONG PASSWORD for user 'root'"	Re-run START_HERE with the correct MySQL root password (or set another user in `config.py`).
"MySQL server is NOT RUNNING"	Windows *Services* → **MySQL80** → Start.
Badge shows "SQLite connected"	MySQL was not reachable, so the app used its built-in copy. The console window shows the reason.
CHECK constraints not enforced	MySQL older than 8.0.16 ignores CHECK. Use 8.0.16 or newer.
Port 5000 busy	Close the other program using it, or change `port=5000` at the end of `app.py`.